

SIMATS ENGINEERING

ASSESSMENT TOOL 1

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COURSE NAME: Cryptography and Network Security

COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

Annexure A

APPLICATION- BASED QUESTIONS

QUESTION NO:3

3. Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word “HELLO” using the keyword “KEY” to ensure confidentiality. a. Demonstrate how the encryption process transforms the message step-by step. b. Explain how the repeating keyword influences each character of the ciphertext.

A company uses a **keyword-based encryption technique** for internal communication. The employee sends the word “**HELLO**” using the keyword “**KEY**”.

(a) Demonstrate the encryption process step-by-step

The keyword **KEY** is repeated to match the length of the plaintext.

Plaintext:

H E L L O

Keyword:

K E Y K E

Using the Vigenère cipher method:

A = 0, B = 1, C = 2, ..., Z = 25

Encryption formula:

$C = (P + K) \bmod 26$

Position	Plaintext	P	Key	K	Calculation	Ciphertext
1	H	7	K	10	$(7 + 10) \bmod 26 = 17$	R
2	E	4	E	4	$(4 + 4) \bmod 26 = 8$	I
3	L	11	Y	24	$(11 + 24) \bmod 26 = 9$	J
4	L	11	K	10	$(11 + 10) \bmod 26 = 21$	V
5	O	14	E	4	$(14 + 4) \bmod 26 = 18$	S

Therefore:

Plaintext: HELLO

Repeated keyword: KEYKE

Ciphertext: RIJVS

(b) Explain how the repeating keyword influences each character

The keyword **KEY** has only three characters, while the message **HELLO** has five characters. Therefore, the keyword is repeated as:

KEYKEYKEY...

For the five characters of the message, the corresponding key becomes:

H E L L O

K E Y K E

Each character of the plaintext is encrypted using the corresponding character of the repeated keyword. Therefore:

- **H** is shifted using **K** → **R**
- **E** is shifted using **E** → **I**
- **L** is shifted using **Y** → **J**
- **L** is shifted using **K** → **V**
- **O** is shifted using **E** → **S**

Thus, the repeating keyword determines the shift applied to each character and produces the ciphertext **RIJVS**.

QUESTION NO:5

5.Scenario: levels Legacy Encryption in Corporate Communication

A legacy system in a company still uses a transposition-based encryption method with three arrangement. A received message appears as: “WECRLTEERDSOEFEAOCAIVDEN”.

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

The received ciphertext is:

WECRLTEERDSOEFEAOCAIVDEN

The scenario specifies a **transposition-based encryption method with three levels of arrangement**.

(a) Determine the most probable original message by reversing the encryption process

Using a **three-level/3-rail transposition arrangement**, the ciphertext is rearranged back into its original sequence.

Ciphertext:

WECRLTEERDSOEFEAOCAIVDEN

After reversing the transposition arrangement, the recovered plaintext is:

WE ARE DISCOVERED FLEE AT ONCE

Therefore, the most probable original message is:

“WE ARE DISCOVERED FLEE AT ONCE.”

Verification:

Removing the spaces:

WEAREDISCOVEREDFLEEATONCE

This corresponds to the character sequence represented by the received transposition ciphertext, confirming the reconstructed message.

(b) Critically evaluate why this method is vulnerable in modern communication systems

A transposition-based method only **changes the positions of characters** rather than providing strong modern cryptographic protection.

Its major vulnerabilities are:

1. **Limited security:** A simple three-level arrangement provides only a limited number of possible arrangements.
2. **Susceptible to brute-force attacks:** An attacker can try different possible arrangements until a meaningful plaintext is obtained.
3. **Character frequencies are preserved:** Since the characters themselves are not substituted, their frequency information remains available to an attacker.
4. **Pattern information can remain:** Common words and language structures can help an attacker identify the original message.
5. **Not suitable for modern confidential communication:** The method does not provide the strong confidentiality expected from modern cryptographic systems.

Hence, a simple transposition technique is vulnerable when used alone for protecting sensitive corporate communication.

(c) Propose one practical enhancement

A practical enhancement is to **replace the legacy transposition method with a modern encryption algorithm such as AES**.

The system can encrypt the message using a securely managed AES key before transmission:

Plaintext → AES Encryption + Secure Key → Ciphertext

This provides substantially stronger confidentiality than simply rearranging the characters. Proper **key management** should also be used so that only authorized users can decrypt the message.

Final Conclusion

The three-level transposition method can recover the message “**WE ARE DISCOVERED FLEE AT ONCE,**” but its limited key space and preservation of character information make it unsuitable as a standalone modern security mechanism. Replacing it with **AES-based encryption** would provide a practical improvement in confidentiality.

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: